



Fig. 7: Remote control of LG TV

cameras. Use a good-quality heat-sink for both the regulators.

Power amplifier

A simple power amplifier (see Fig. 5) is used to amplify the sound signal from the cameras and hear it on a small speaker.

The audio power amplifier is built around IC UTC820 (IC5). UTC820 is a monolithic integrated circuit audio amplifier delivering an output of 1.2 watts at 9V on an 8-ohm speaker load with 10 per cent total harmonic distortion (THD) and good ripple rejection. It is designed for audio-frequency class-B amplifier with a wide operating

supply voltage range of 3V to 14V and minimum external components.

The pin configuration of IC UTC820 is shown in Fig. 6. Components at pin 2 of the IC5 decide closed-loop voltage gain (dB) of the amplifier. Here a 47-ohm resistor (R11) and a 22μF capacitor (C12) are used and, with these values, about 45dB closed-loop voltage gain is achieved. You can reduce the component values for more gain but it's not required here as maximum gain of 75 dB is possible. Capacitor C10 (100μF) connected between pins 5 and 7 of IC5 is used for bootstrap and improving the low-frequency signal. The amplifier output at pin 5 is connected to the speaker via C13. The audio signal from the cameras is fed to input pin 3 of the IC via volume control VR2 (10-kilo-ohm), capacitor C17 and resistor R13. Here the output power is limited to about 500 mW at 8-ohm load approximately, so use of a good-quality 6.4cm speaker is recommended.

Functions of remote control keys

The functions of various remote keys are shown in Fig. 7.

Power on/off key. Press it once to switch on the switcher and press again to turn off or keep the switcher in standby mode and vice versa. (AC mains should be 'on' in both the cases.)

Camera selection key. Select any of the six cameras manually by pressing any of the numerical keys '1' through '6' once. For example, to select camera-3 press '3' on the remote, and to select camera-5 press '5' on the remote.

Auto/manual. By default, the microcontroller sets the mode of selection to manual. (In that case, you have

to select the cameras by pressing the number keys or the camera up/down keys.) To change to automatic selection (auto) mode, press 'Mute' key and the switcher selects the cameras one by one automatically. Pressing this key again will return the mode of selection to manual, and vice versa.

Camera up key (PR Δ key). By pressing the camera-up key, the switcher selects the next camera. For example, if the LCD is showing camera-2, pressing this key once makes the switcher jump to camera-3. Each successive pressing of this key will make the switcher jump to the next camera.

Camera down key (PR ∇ key). By pressing the camera-down key, the switcher selects the preceding camera. For example, if the LCD is showing camera-4, pressing this key once makes the switcher jump back to camera-3. On each successive pressing of this key, the switcher jumps back to the previous camera.

Time set up/down key (Vol+/Vol- key). In auto mode, the switcher selects the cameras one by one and the picture of the selected camera is held up for a certain time. By default, the microcontroller sets this hold time as 5 seconds. You can change this time for each camera individually from 0 to 255 seconds.

To change this time, press the edit key (TV/AV) once. The switcher goes to the edit mode. Select the camera (for which you want to change the time) by pressing the camera up/down key. On each successive pressing of the time set up key, the time increments up to 255 seconds maximum.

On each successive pressing of the time set-down key, the time decrements up to zero. If you set 0-second time for any camera, the switcher bypasses this camera during selection in auto mode. For example, if you set the time as 0-second for camera-3, in auto mode, the switcher selects camera-1, camera-2, camera-4, camera-5 and camera-6. Here the switcher bypasses camera-3. This function is useful if you